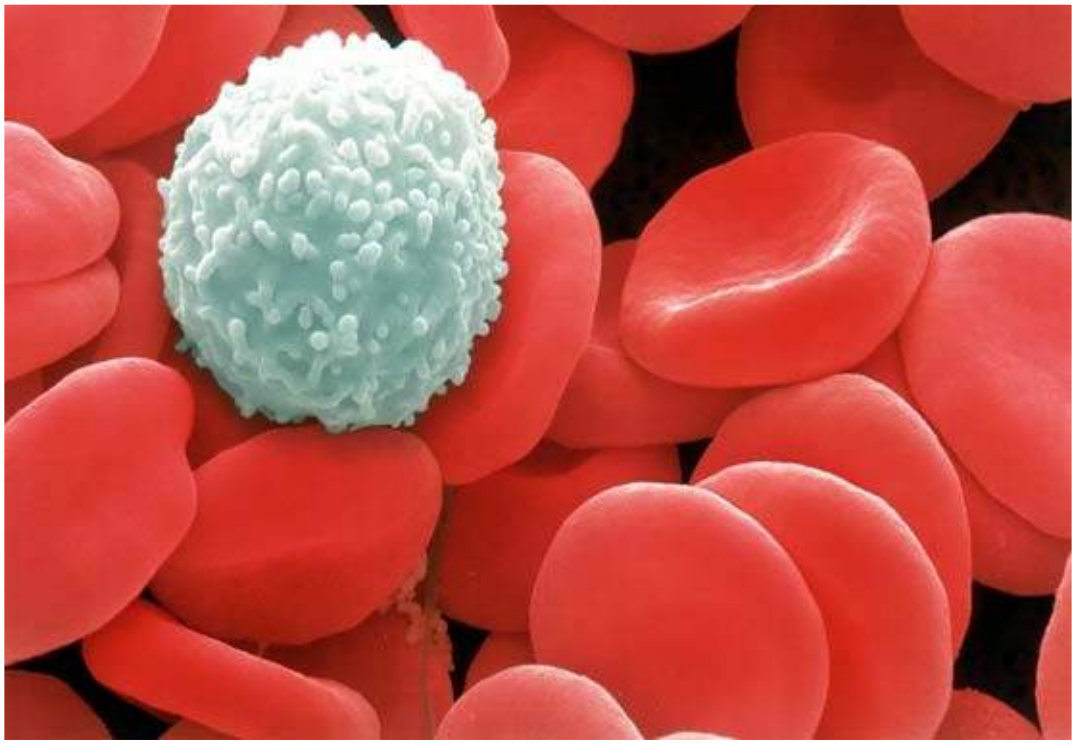
Immunity

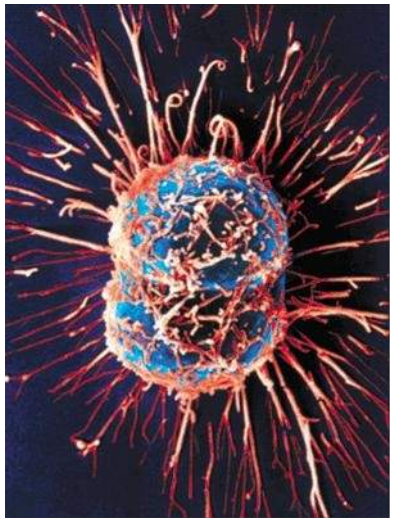


Video

What is the immune system?

- The body's defense against disease causing organisms, malfunctioning cells, and foreign particles







Innate (Non-Specific) Immunity

- Innate immunity is the inborn capacity of the body to resist pathogens. By chance, if the organisms enter the body, innate immunity eliminates them before the development of any disease.

Structures and Mediators	Mechanism
Gastrointestinal tract	Enzymes in digestive juices and the acid in stomach destroy the toxic substances or organisms entering digestive tract through food Lysozyme present in saliva destroys bacteria
Respiratory system	Defensins and cathelicidins in epithelial cells of air passage are antimicrobial peptides, Neutrophils, lymphocytes, macrophages and natural killer cells present in lungs act against bacteria and virus
Skin	The keratinized stratum corneum of epidermis protects the skin against toxic chemicals. The β-defensins in skin are antimicrobial peptides, Lysozyme secreted in skin destroys bacteria
Urinogenital system	Acidity in urine and vaginal fluid destroy the bacteria
Phagocytic cells	Neutrophils, monocytes and macrophages ingest and destroy the microorganisms and foreign bodies by phagocytosis
Interferons	Inhibit multiplication of viruses, parasites and cancer cells
Complement proteins	Accelerate the destruction of microorganisms

ACQUIRED IMMUNITY OR SPECIFIC IMMUNITY

- Acquired immunity is the resistance developed in the body against any specific foreign body like bacteria, viruses, toxins, vaccines or transplanted tissues. So, this type of immunity is also known as specific immunity.

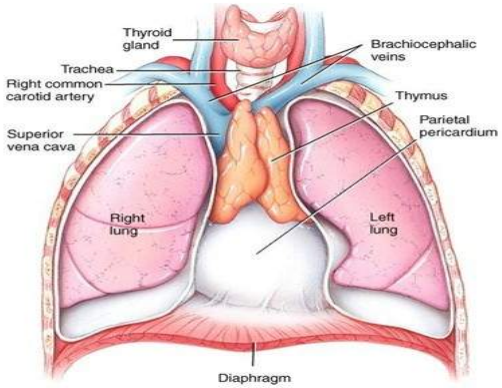
Types of Acquired Immunity

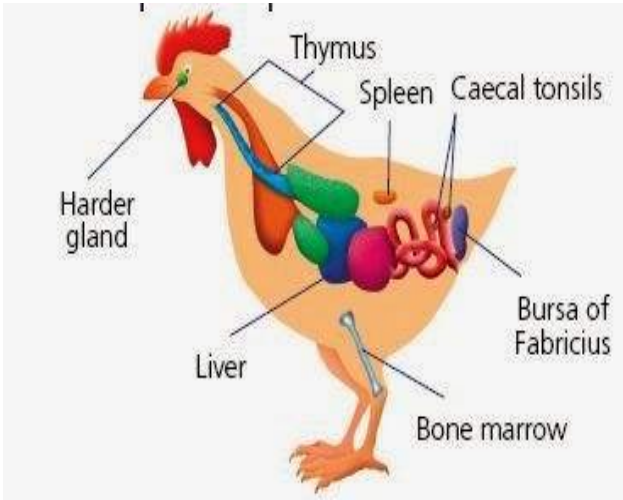
- Cellular immunity (T cell mediated)
- Humoral immunity. (B cell mediated)

- Lymphocytes are responsible for the development of these two types of immunity.

Types of lymphocytes

- **T- lymphocytes:** T lymphocytes are processed in thymus. The processing occurs mostly during the period between just before birth and few months after birth.
- **B – lymphocytes :** B lymphocytes were first discovered in the bursa of Fabricius in birds, hence the name B lymphocytes.





Types of T Lymphocytes

- **Helper T cells or inducer T cells.** These cells are also called **CD4 cells** because of the presence of molecules called CD4 on their surface.
- **Cytotoxic T cells or killer T cells.** These cells are also called **CD8 cells** because of the presence of molecules called CD8 on their surface.
- **Suppressor T cells.**
- **Memory T cells.**

Storage of T Lymphocytes

- After the transformation, T lymphocytes are stored in,
- Lymphoid tissues of lymph nodes
- Spleen
- Bone marrow
- GI tract.

DEVELOPMENT OF CELL-MEDIATED IMMUNITY

- The immunity developed by cell-mediated response. Cellular immunity is the major defense mechanism against infections by viruses, fungi and few bacteria like tubercle bacillus.
- It involves several types of cells such as T lymphocytes, macrophages and natural killer cells and hence the name cell mediated immunity.

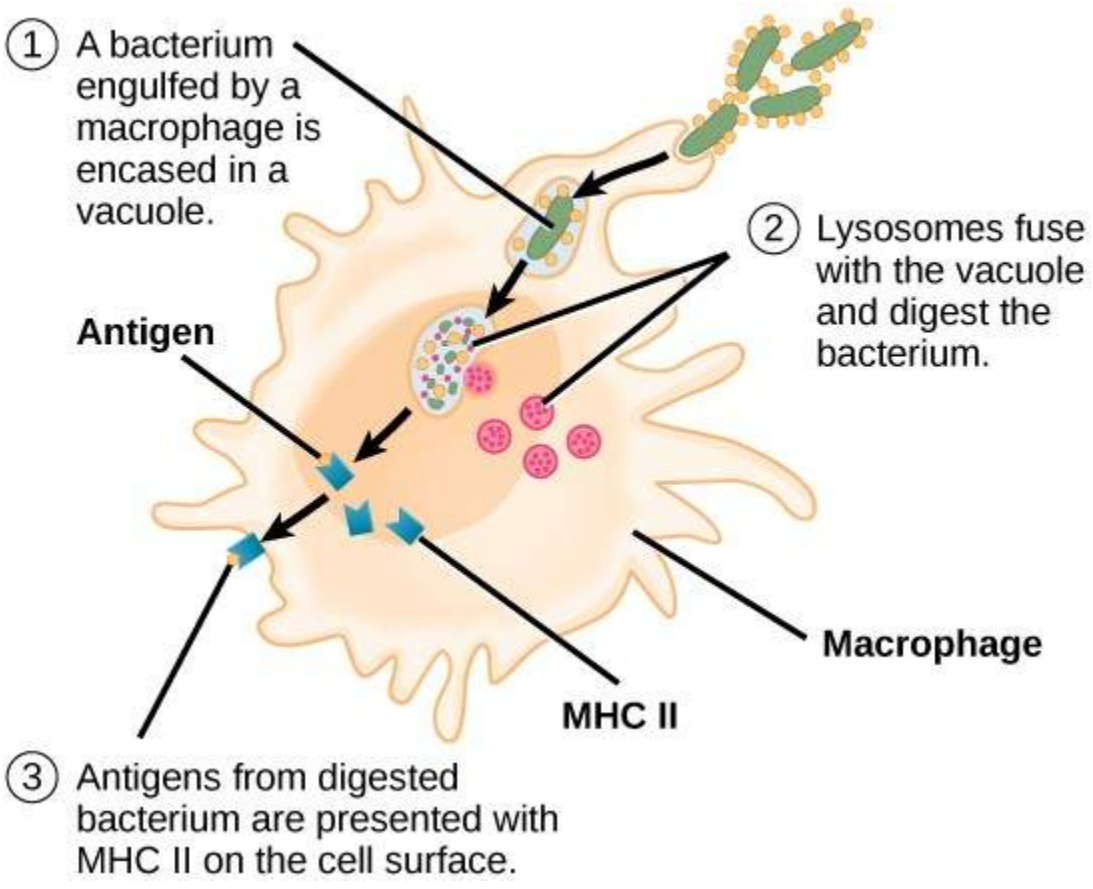
- Usually, the invading microbial or non-microbial organisms carry the antigenic materials.
- These antigenic materials are released from invading organisms and are presented to the helper T cells by antigen-presenting cells

- Antigen-presenting cells are of three types:
- **1. Macrophages**
- **2. Dendritic cells**
- **3. B lymphocytes.**
- Among these cells, macrophages are the major antigen- presenting cells

Role of Antigen-presenting Cells

- After phagocytosis, the antigen from these organisms is digested into small peptide products. These antigenic peptide products move towards the surface of the antigen-presenting cells and bind with human leukocyte antigen (HLA).

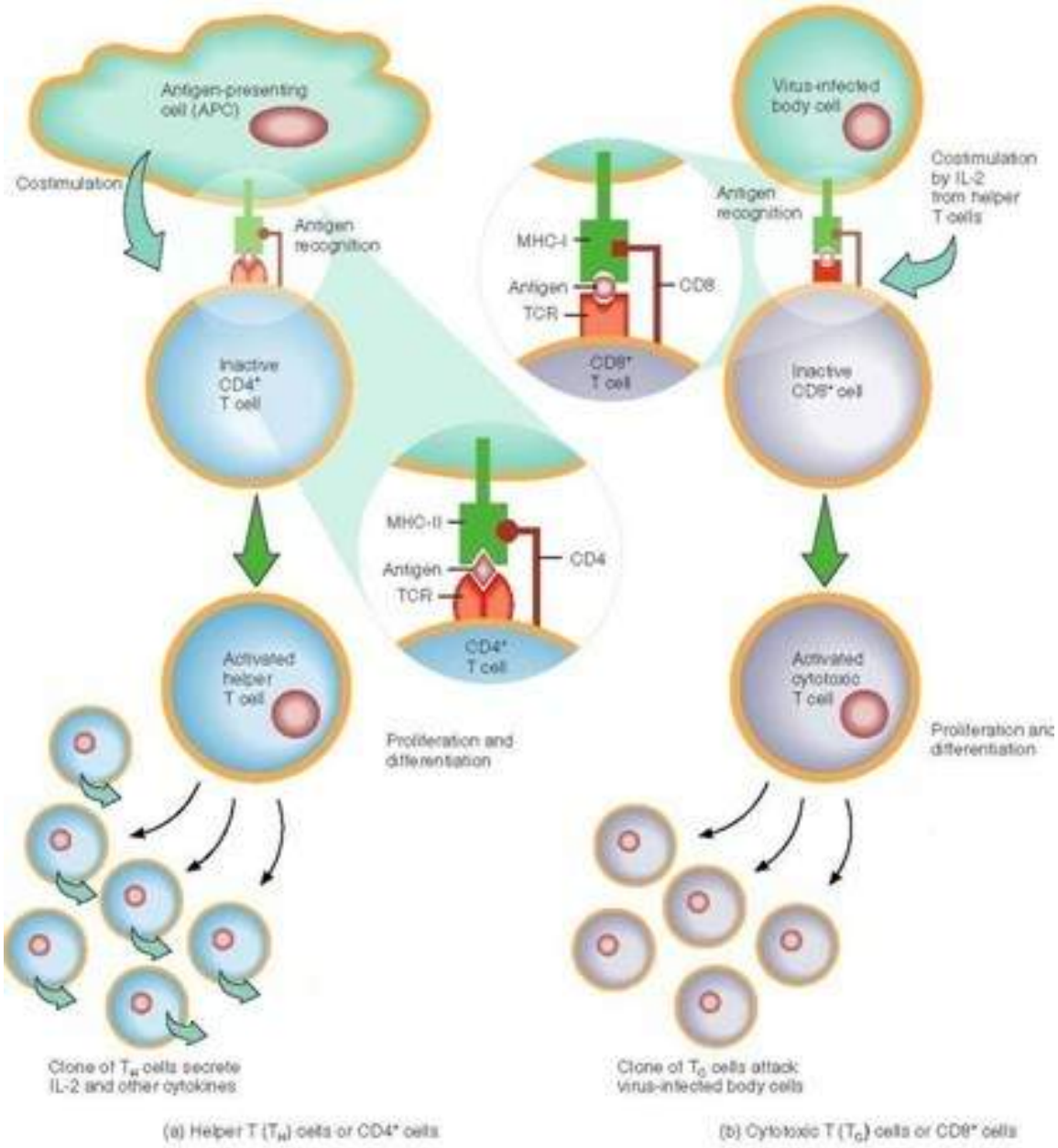
Antigen presenting cell



Presentation of Antigen

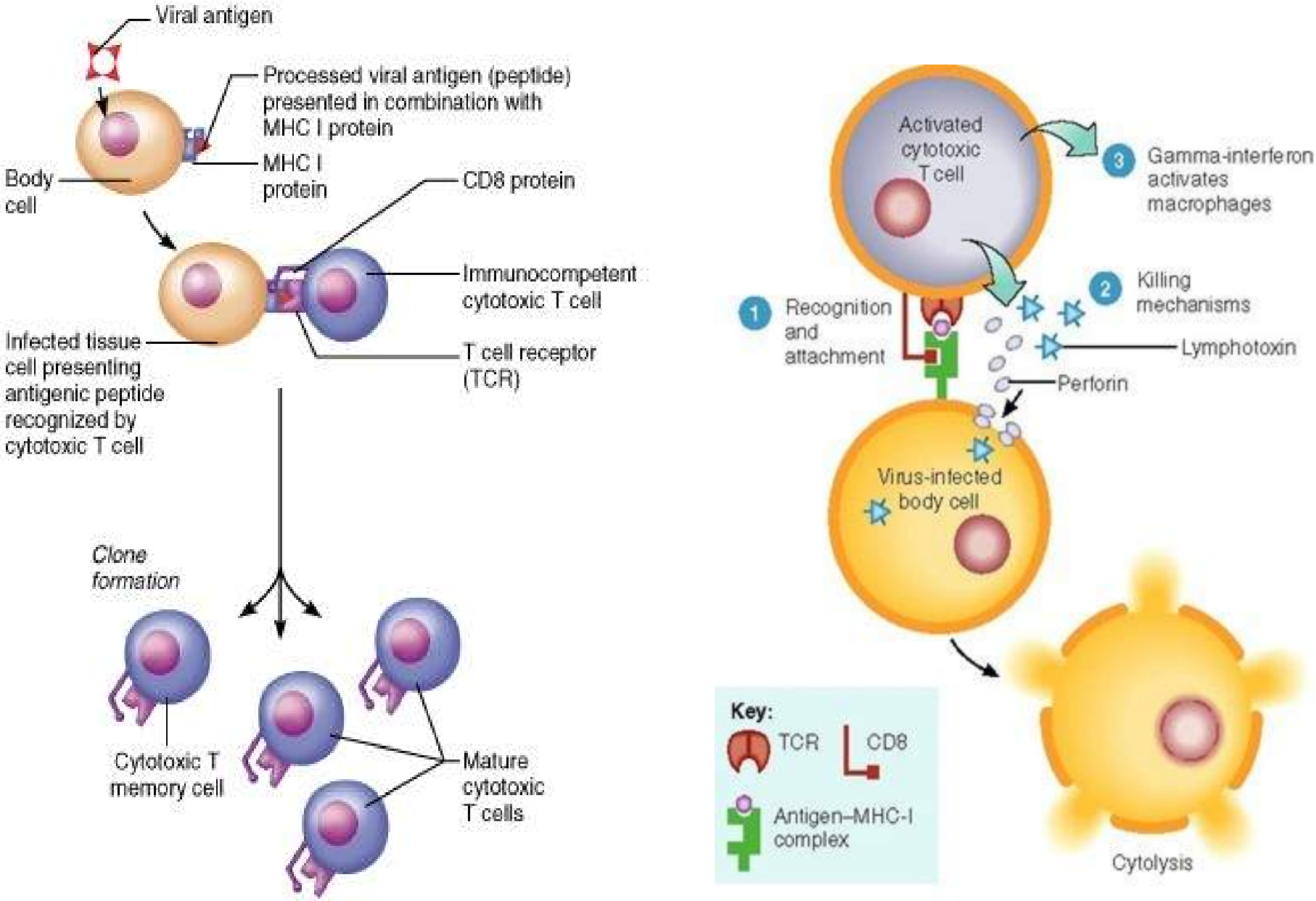
- Helper T cell recognizes the antigen displayed on the surface of the antigen-presenting cell with the help of its own surface receptor protein called T cell receptor.
- Interaction b/w helper T cell receptor and antigen.
- Activation of helper T cells
- Interleukin-1 released from macrophages facilitates the activation and proliferation of helper T cells.

- Proliferated helper T cells enter the circulation for further actions. IL2 secreted by these cells stimulate other T helper cells
- Simultaneously, the antigen which is bound to class II
- MHC molecules activates the B cells also. Development of humoral immnity


Antigen-presenting cell (APC)
Costimulation
Antigen recognition
Inactive CD4+ T cell
MHC-II
Antigen
TCR
CD4
CD4+ T cell
Activated helper T cell
Proliferation and differentiation
Clone of T_H cells secrete IL-2 and other cytokines
(a) Helper T (T_H) cells or CD4+ cells
Virus-infected body cell
Antigen recognition
Costimulation by IL-2 from helper T cells
MHC-I
Antigen
TCR
CD8
CD8+ T cell
Inactive CD8+ cell
Activated cytotoxic T cell
Proliferation and differentiation
Clone of T_C cells attack virus-infected body cells
(b) Cytotoxic T (T_C) cells or CD8+ cells

Cytotoxic T cell activation

- Activation of CD8 cells by CD4 cells
- Receptors located on the outer surface of Cytotoxic cells bind the antigens or organisms tightly with cytotoxic T cells.
- cytotoxic T cells enlarge and release cytotoxic substances like the lysosomal enzymes.
- Destruction of invading organism



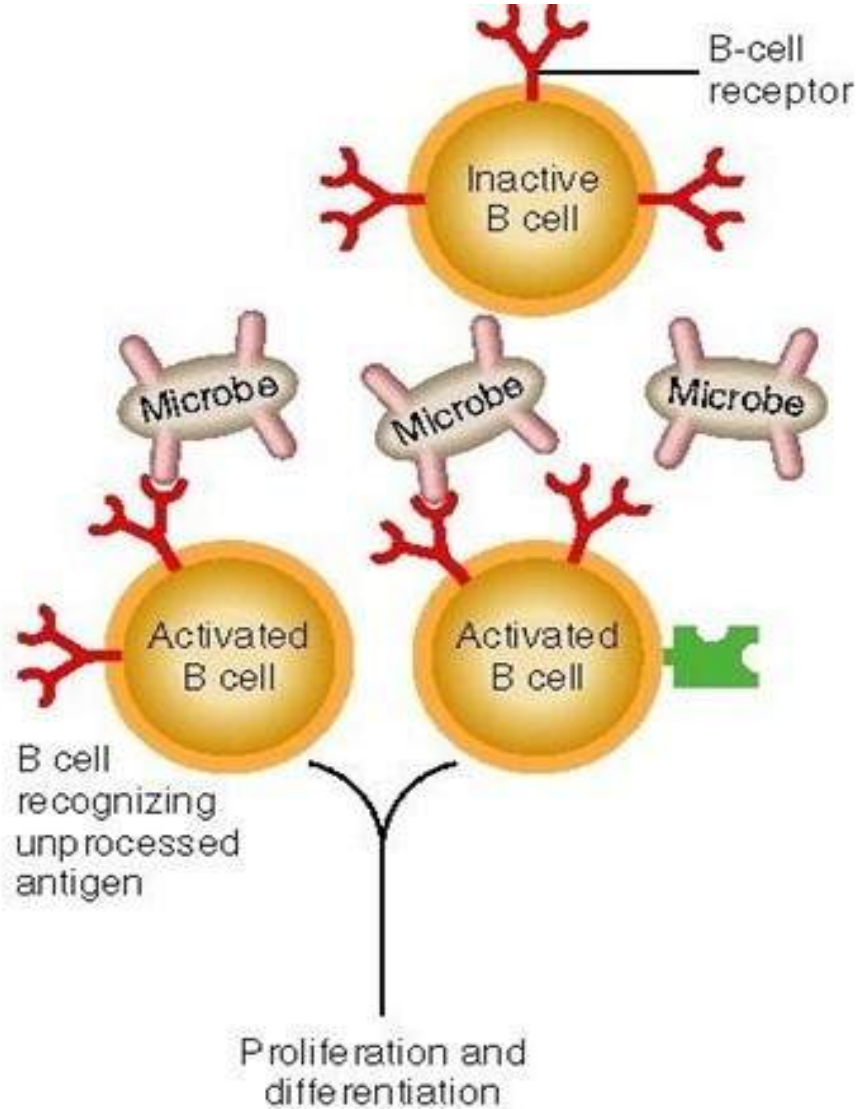
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- **Cytotoxic T cells** also destroy cancer cells, transplanted cells, such as those of transplanted heart or kidney or any other cells, which are foreign bodies.
- **ROLE OF SUPPRESSOR T CELLS:**
- suppress the activities of the killer T cells.
- Play an important role in preventing the killer T cells from destroying the body's own tissues along with invaded organisms.
- suppress the activities of helper T cells

- **Memory T cells:** Some of the T cells activated by an antigen do not enter the circulation but remain in lymphoid tissue. These T cells are called memory T cells.
- When the body is exposed to the same organism for the second time, the memory cells identify the organism and immediately activate the other T cells.

DEVELOPMENT OF HUMORAL IMMUNITY

- immunity mediated by antibodies, which are secreted by B lymphocytes.
- B lymphocytes secrete the antibodies into the blood and lymph.



- Antigen-presenting cells present the antigenic products bound with HLA (which is present in class II MHC molecule) to B cells.

Activation of B cells

- B cells recognizes the antigen by its receptor.
- Receptor and the antigen. This reaction activates B cells.
- interleukin-1, which facilitates the activation and proliferation of B cells.
- Activated B cells proliferate and the proliferated cells carry out the further actions.
- Proliferated B cells are transformed into two types of cells:
- 1. Plasma cells
- 2. Memory cells.

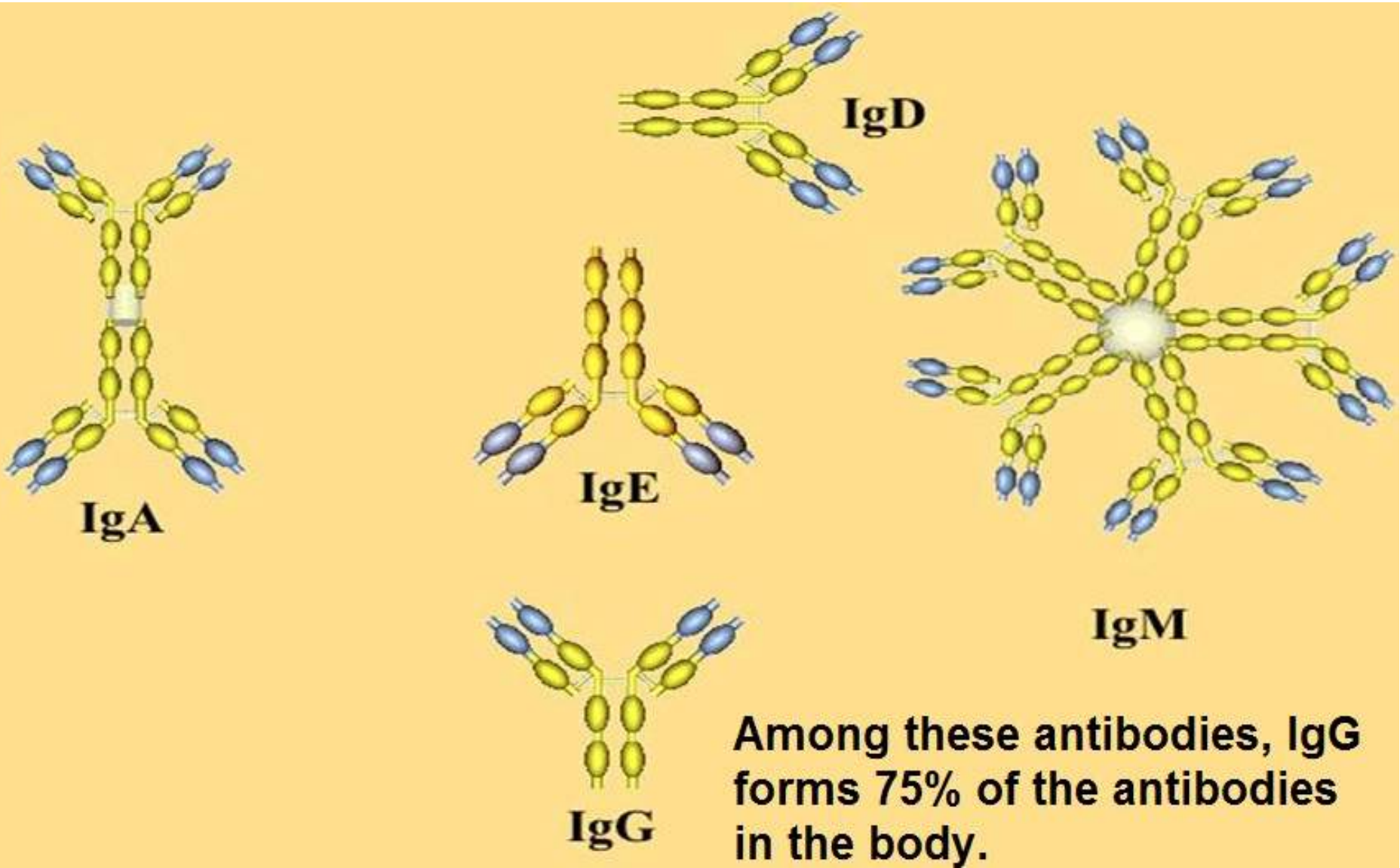
Plasma cells

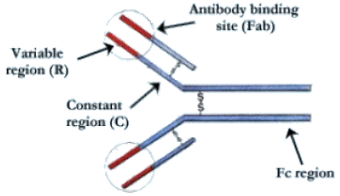
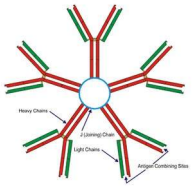
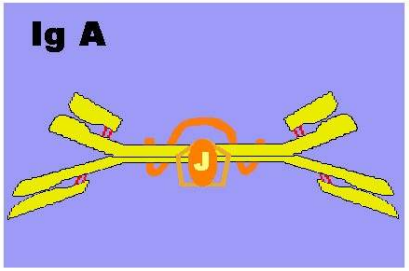
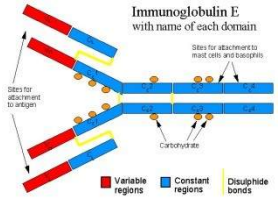
- Plasma cells destroy the foreign organisms by producing the antibodies. each plasma cell produces about 2000 molecules of antibodies per second.

- **Antibodies:** An antibody is a protein that produced by B lymphocytes in response to the presence of an antigen.

- Antibody is gamma globulin in nature and it is also called immunoglobulin (Ig). Immunoglobulins form 20% of the total plasma proteins. Antibodies enter almost all the tissues of the body.

Different Immunoglobulins



Type	Number of ag binding sites	Site of action	Functions
IgG 	2	•Blood •Tissue fluid •CAN CROSS PLACENTA	•Increase macrophage activity •Antitoxins •Agglutination
IgM 	10	•Blood •Tissue fluid	Agglutination
IgA 	2 or 4	•Secretions (saliva, tears, small intestine, vaginal, prostate, nasal, breast milk)	•Stop bacteria adhering to host cells •Prevents bacteria forming colonies on mucous membranes
IgE 	2	Tissues	•Activate mast cells → HISTAMINE •Worm response

- **ROLE OF MEMORY B CELLS:** The memory cells are in inactive condition until the body is exposed to the same organism for the second time.
- **ROLE OF HELPER T CELLS:** Helper T cells are simultaneously activated by antigen.
- Activated helper T cells secrete two substances called interleukin-2 and B cell growth factor, which promote:
- 1. Activation of more number of B lymphocytes.
- 2. Proliferation of plasma cells.
- 3. Production of antibodies.

Mechanism of Actions of Antibodies

Direct Actions of Antibodies:

- Agglutination:
- Precipitation:
- Neutralization:
- Lysis:

Actions of Antibodies through Complement System

- It is a system of plasma enzymes, which are identified by numbers from C1 to C9. Including the three subunits of C1 (C1q C1r C1s), there are 11 enzymes in total. Normally, these enzymes are ininactive form and are activated in following ways:
- a. Classical pathway
- b. Alternate pathway.

Classical pathway

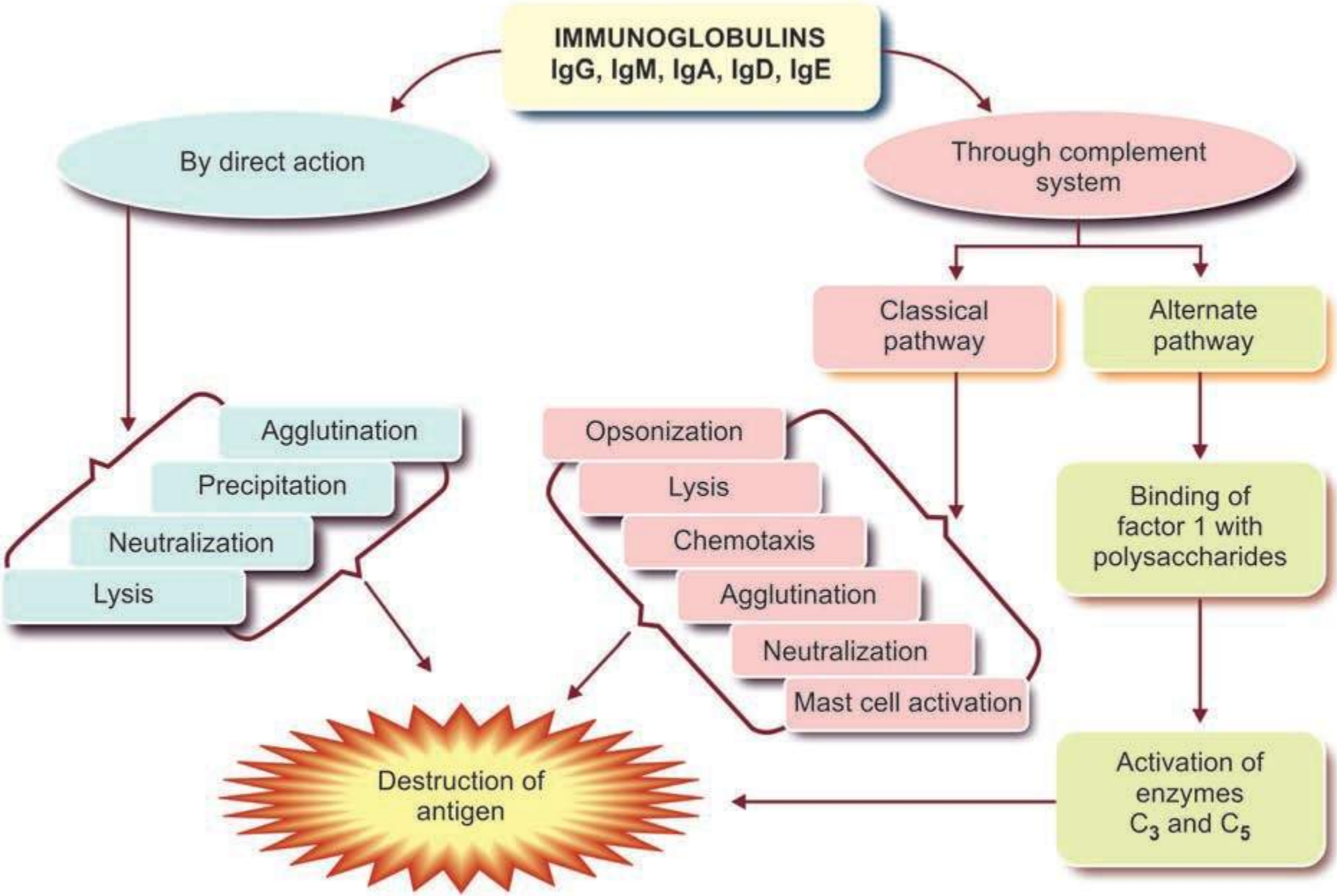
- In this the C1 binds with the antibodies and triggers a series of events in which other enzymes are activated in sequence. Perform following functions.
- ***Opsonization:*** Activation of neutrophils and macrophages to engulf the bacteria, which are bound
- with a protein in the plasma called opsonin.
- **ii. *Lysis:*** Destruction of bacteria by rupturing the cell membrane.
- **iii. *Chemotaxis***: Attraction of leukocytes to the site of antigen-antibody reaction.

- **iv. *Agglutination***: Clumping of foreign bodies like RBCs or bacteria.
- **v. *Neutralization***: Covering the toxic sites of antigenic products.
- **vi. *Activation of mast cells and basophils***, *which liberate histamine*: Histamine dilates the blood vessels and increases capillary permeability.
- So, plasma proteins from blood enter the tissues and inactivate the antigenic products.

Alternate pathway

- Complementary system is also activated by another way, which is called alternate pathway. It is due to a protein in circulation called factor I. It binds with polysaccharides present in the cell membrane of the invading organisms.

- This binding activates C3 and C5, which ultimately attack the antigenic products of invading organism


IMMUNOGLOBULINS
IgG, IgM, IgA, IgD, IgE
By direct action
Through complement system
Classical pathway
Alternate pathway
Agglutination
Precipitation
Neutralization
Lysis
Opsonization
Lysis
Chemotaxis
Agglutination
Neutralization
Mast cell activation
Binding of factor 1 with polysaccharides
Destruction of antigen
Activation of enzymes C_3 and C_5

NATURAL KILLER CELL

- Natural killer (NK) cell is a large granular cell that plays an important role in defense mechanism of the body. It has an indented nucleus.
- Destroys the viruses
- 2. Destroys the viral infected or damaged cells, which might form tumors
- 3. Destroys the malignant cells and prevents development of cancerous tumors.
- Secretes cytokines such as interleukin-2, interferons,
- colony stimulating factor (GM-CSF) and tumor necrosis factor-α.

THANKS